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RESEARCH ARTICLE

Defect equilibria and thermophysical properties of CeO_{2-x} based on experimental data and density functional theory calculation result

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Abstract

Properties of CeO_2 were evaluated by density functional theory (DFT) simulation to determine bandgap, Frenkel defect formation energy, and defect migration energy. Bandgap and Frenkel defect formation energy were used to analyze defect equilibria. Oxygen partial pressure dependence of defect equilibria was evaluated based on oxygen potential experimental data and DFT calculation, and a Brouwer diagram was derived. The defect formation energies, including Frenkel defect, electron-hole pair, and so on, were determined and used to evaluate the properties, including oxygen diffusion coefficients, electrical conduction, heat capacity, and thermal conductivity. Mechanisms of various properties were discussed for a deeper understanding based on defect chemistry, and the relationship among properties was systematically described.

1 | INTRODUCTION

Cerium dioxide has been studied for the development of catalyst and electricity storage materials.[1] In research and development for advanced nuclear fuels, cerium oxide has been used as a surrogate material for PuO_2 in various experiments because both materials have very similar mechanical and thermal physical properties.[2–5] Both oxides of CeO_2 and PuO_2 have a fluorite crystal structure and are nonstoichiometric compounds that are stable over the hypo-stoichiometric composition range. In addition, chemical, thermal and mechanical properties are very similar in both materials. A significant difference in the properties of both materials is their electrical conduction mechanisms at high temperatures. CeO_2 has an ionic conduction mechanism near stoichiometric composition, and its mechanism changes to electronic conduction contribution in hypo-stoichiometric composition.[6–9] On the other hand, PuO_2 is electronic conduction in all compositions.[10] The electric conduction mechanism is determined by the type of lattice defect which is dominant at high temperatures. These materials have similar properties, despite their completely different electrical conduction mechanisms. In order to gain a deeper understanding of the basic properties of these materials, it is of great importance to understand the relationship between the lattice defect and properties in both materials.

CeO_2 has a fluorite structure and is an oxygen nonstoichiometric compound that is stable over a wide hypo-stoichiometric composition range as PuO_2. Stoichiometry significantly affects various properties such as thermal conductivity, electrical conductivity, and so on. Recently, Suzuki et al.[5] have systematically measured the mechanical and thermal properties of CeO_{2-x}. In their paper, oxygen chemical potential data were measured and analyzed based on defect chemistry. Despite attempts to understand property change in the nonstoichiometric composition, the relationship between defect concentration and basic

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properties in CeO_{2-x} has not yet been quantitatively evaluated.

Kato et al. reported defect chemistry and the basic properties of PuO_2.[11] They constructed a Brouwer diagram from experimental oxygen potential data and reported that there is the same relationship between the deviation x, the electrical conductivity, and the oxygen diffusion coefficient depending on the oxygen partial pressure (P_{O_2}) in PuO_{2-x}. In addition, heat capacity and thermal conductivity were also evaluated. In this report, a Brouwer diagram for CeO_{2-x} was constructed and the relationship between lattice defect and properties, including deviation x, electrical conductivity, and oxygen diffusion coefficient, heat capacity, and thermal conductivity, was evaluated.

2 | CALCULATION METHODS

2.1 | Density functional theory

The analyses of defect equilibria and thermophysical properties need various physical parameters, such as defect formation energies, bandgap, elastic constants, defect migration energies, and so on. In this work, most of these parameters are evaluated by density functional theory (DFT). The calculation package adopted was Vienna Ab initio Simulation Package (VASP),[12] which supports the projector augmented wave method.[13,14] In these calculations, two types of exchange-correlation energy: Perdew-Burke-Ernzerhof revised for solids (PBEsol)[15] and Strongly Constrained and Appropriately Normed (SCAN),[16] were used. The former was a generalized gradient approximation (GGA) and the latter was metaGGA. Through this work, the energy cutoff was fixed to 500 eV.

The lattice constants for the optimized structures were obtained using the primitive cell of fluorite structure with k-points $15 \times 15 \times 15$. Using the optimized structure, elastic stiffness constants were evaluated from the stress tensors with finite strains. The bulk, shear, and Young's moduli and Poisson's ratio were calculated by Hill's formalism[17] using the elastic stiffness constants.

2.2 | Defect formation and migration energies

The defect formation and migration energies were calculated with $2 \times 2 \times 2$ supercells which contain 96 atoms for stoichiometric $Ce_{32}O_{64}$. In the super-cell calculations, k-points were set to $2 \times 2 \times 2$. The lattice constants used in those calculations were fixed to those of optimized structures. For the defect formation energy, it was assumed that the charges of oxygen vacancy and oxygen interstitial were +2 and −2, respectively, since the ionic charge of oxygen in CeO_2 is expected to be −2. The charges of defects were controlled by increasing or decreasing the total number of electrons in the system. Thus, oxygen Frenkel defect formation energy can be described as

$$E_{\text{Frenkel}} = E\left(Ce_{32}O_{63} - 2e\right) + E\left(Ce_{32}O_{65} + 2e\right) - 2E\left(Ce_{32}O_{64}\right), \quad (1)$$

where $E(Ce_{32}O_{64+x})$ is the energy of the relaxed structure of $Ce_{32}O_{64+x}$ obtained by DFT, and $-2e$ means that the number of electrons is reduced by 2.

The migration energies were calculated by the nudged elastic band method[18] with 5 intermediate states for +2-charged oxygen vacancy and −2-charged oxygen interstitial. In these calculations, we adopted only SCAN exchange-correlation energy, since SCAN reproduced lattice constants and elastic constants better than PBEsol as seen below. In the case of oxygen vacancy migration, the minimum energy path is the direct jump of oxygen vacancy to the nearest oxygen site. On the minimum energy path of oxygen interstitial, the oxygen interstitial pushes out the nearest oxygen atom, and the pushed atom moves to another interstitial site.

3 | RESULTS AND DISCUSSION

3.1 | Density functional theory

The calculated values using DFT are summarized in Table 1. Experimental data of lattice parameters and mechanical properties are also shown for comparison. The simulation with SCAN exchange-correlation energy described the experimental data very well. Therefore, the simulation with SCAN exchange-correlation energy was employed in the current analysis.

The calculated values using DFT were used to evaluate the thermophysical properties. First, defect equilibria and defect concentration were evaluated by constructing a Brouwer diagram using the calculation results of bandgap and Frenkel defect formation energies, and experimental data of oxygen potential. The Brouwer diagram could represent point defect concentration as functions of temperature and P_{O_2}. Subsequently, oxygen diffusion coefficient, electrical conduction, heat capacity, and thermal conductivity were evaluated. In the evaluation of heat capacity and thermal conductivity at high temperatures, the contribution of defect formation was considered.

3.2 | Brouwer diagram construction

In nonstoichiometric compounds, it is well known that various physical properties are closely related to the nature

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TABLE 1 Density functional theory (DFT) calculation results of CeO_2

Exchange-correlation energy	DFT PBEsol	DFT SCAN	Experiment[3,5]
Band gap	1.92 eV (185 kJ/mol)	2.2 eV (212 kJ/mol)	—
Oxygen Frenkel defect formation energy	4.0 eV (386 kJ/mol)	4.28 eV (413 kJ /mol)	—
Oxygen vacancy migration energy	—	0.557 eV (53.7 kJ/mol)	—
Interstitial oxygen migration energy	—	0.798 eV (77.0 kJ/mol)	—
Lattice parameter (Å)	5.397	5.420	5.4418
Young modulus (GPa)	219	222	228
Shear modulus (GPa)	83	84	86
Poisson's ratio	0.315	0.317	0.318
Bulk modulus (GPa)	197	202	208
C_{11} (GPa)	366	380	—
C_{12} (GPa)	112	113	—
C_{44} (GPa)	63	61	—
Longitudinal sound speed, V_l (m/s)	—	3420.3	—
Transverse sound speed, V_t (m/s)	—	6612.9	—
Debye temperature, T_D (K)	—	489	—
Grüneisen constant, γ	—	2.96	—
Coefficient of thermal expansion, α (K^{-1})	—	—	3.39×10^{-5}

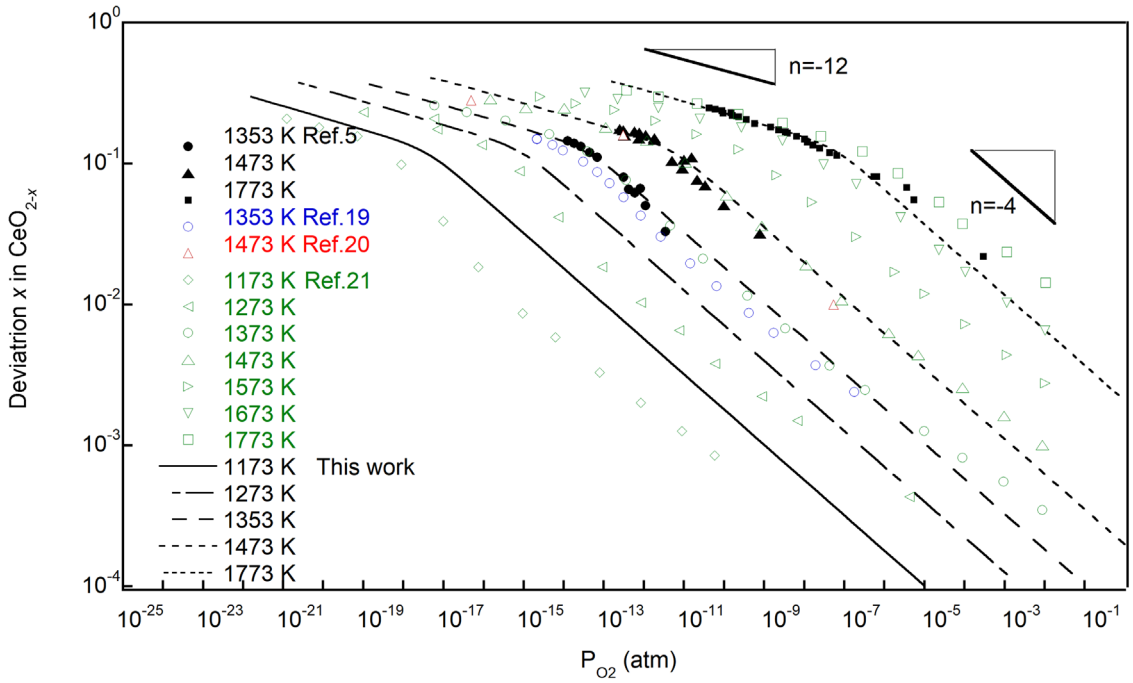

FIGURE 1 Relationship between deviation x in CeO_{2-x} and oxygen partial pressure

of defect reactions. Therefore, defect equilibria in CeO_{2-x} were evaluated, and then the Brouwer diagram was constructed as a first analysis.

Suzuki et al. determined the oxygen potential ($\Delta\overline{G_{O_2}}$) of CeO_2 as a function of temperature and deviation x in CeO_{2-x}.[5] Figure 1 shows the dependence of deviation x on P_{O_2}.[5,19,20] In this analysis, it is well-known that x is proportional to $P_{O2}{}^{1/n}$. The regions of $n = -4$ and –12 are observed with P_{O_2} in the figure. Suzuki et al.[5] assumed the following defect reactions in each region.

$$O_O^x + Ce_{Ce}^x \rightarrow \left(Vo^{\cdot\cdot}e'\right)^{\cdot} + e' + \frac{1}{2}O_2, \text{for } n = -4 \quad (2)$$

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$$3O_O^x + 5Ce_{Ce}^x \rightarrow (2Vo^{\cdot\cdot}Vo^{\cdot}2Oi)^{\cdots\cdot} + 5e' + \frac{1}{2}O_2, \text{for } n = -12 \quad (3)$$

Equilibrium constants are $K_{n=-4}$ and $K_{n=-12}$, respectively, in defect reaction (1) and (2), oxygen vacancy concentration $[Vo^{\cdot\cdot}]$ can be described by Equations (4) and (5).

$$[Vo^{\cdot\cdot}]_{n=-4} = \left[\left(Vo^{\cdot\cdot}e'\right)^{\cdot}\right] = (K_{n=-4})^{1/2} P_{O2}^{-1/4} \quad (4)$$

$$[Vo^{\cdot\cdot}]_{n=-12} = 1/3\left[(2Vo^{\cdot\cdot}Vo^{\cdot}2Oi)^{\cdots\cdot}\right] = 1/3(K_{n=-12})^{1/6} P_{O2}^{-1/12} \quad (5)$$

$x = [Vo^{\cdot\cdot}]$ was assumed. Suzuki et al. reported that Equations (6) and (7) were obtained by fitting the experimental data with Equations (4) and (5) and derived Equation (8) to represent the deviation x as a function of temperature and P_{O_2}.[5]

$$K_{n=-4} = 45\,000 exp\left(-\frac{340\,000}{RT}\right) \quad (6)$$

$$K_{n=-12} = 14\,702 exp\left(-\frac{345\,000}{RT}\right) \quad (7)$$

$$x = \left[[Vo^{\cdot\cdot}]_{n=-4}^{-8} + [Vo^{\cdot\cdot}]_{n=-12}^{-8} + (0.5)^{-8}\right]^{-1/8} \quad (8)$$

The calculation results are shown with lines in Figure 1 as functions of temperature and P_{O_2} and agree with the experimental data very well.

Panlener et al. investigated $\Delta\overline{G_{O_2}}$ of CeO_2 over the wide temperature region of 1023–1773 K, whose data are plotted in Figure 1.[21] They evaluated n as –5 in the near stoichiometric region; however, the figure showed that n was almost the same as the previous work by Suzuki et al.,[5] which concluded n = –4. The calculation result was consistent with the data reported by Panlener et al. [21] in the region of n = –12 and overestimated in the region of n = –4. This slight difference may be due to measurement error in P_{O_2}.

Now, the following defect reactions (9)–(12) are considered in near-stoichiometric composition.

$$O_O^x \rightarrow Vo^{\cdot\cdot} + 2e' + \frac{1}{2}O_2 \quad (9)$$

$$\frac{1}{2}O_2 \rightarrow Oi'' + 2h^{\cdot} \quad (10)$$

$$null \rightarrow e' + h^{\cdot} \quad (11)$$

$$O_O^x \rightarrow Vo^{\cdot\cdot} + Oi'' \quad (12)$$

The equilibrium constants in each defect reaction can be described by Equations (13)–(16).

$$K_V = [Vo^{\cdot\cdot}]\left[e'\right]^2 P_{O2}^{1/2} \quad (13)$$

$$K_O = \left[Oi''\right]\left[h^{\cdot}\right]^2 P_{O2}^{-1/2} \quad (14)$$

$$K_i = \left[e'\right]\left[h^{\cdot}\right] \quad (15)$$

$$K_F = [Vo^{\cdot\cdot}]\left[Oi''\right] \quad (16)$$

Two types of Brouwer diagram are proposed, in which electrical or Frenkel defects are dominant in near-stoichiometric composition.[11,22,23] Measurement of electrical conductivity showed that CeO_2 was an ionic conductor near stoichiometric composition. Therefore, Frenkel defect is dominant in near-stoichiometric composition, and Equation (17) holds.

$$[Vo^{\cdot\cdot}] = \left[Oi''\right] \quad (17)$$

The following equations were obtained from Equations (13)–(17).

$$[Vo^{\cdot\cdot}]_{Sto.} = \left[Oi''\right]_{Sto.} = {K_F}^{1/2} \quad (18)$$

$$\left[e'\right]_{Sto.} = {K_V}^{1/2}\,{K_F}^{-1/4} P_{O2}^{1/4} \quad (19)$$

$$[h^{\cdot}]_{Sto.} = {K_O}^{1/2}\,{K_F}^{-1/4} P_{O2}^{-1/4} \quad (20)$$

$$K_i = {K_V}^{1/2}\,{K_O}^{1/2}{K_F}^{-1/2} \quad (21)$$

The equilibrium constant was expressed by,

$$K_m = \exp\left(\frac{\Delta S_m}{R}\right) \times \exp\left(-\Delta H_m/RT\right) \quad (22)$$

In the region of n = –4, $[Vo^{\cdot\cdot}]$ is equal to $2[e']$, and Equation (23) was obtained.

$$\left[e'\right]_{n=-4} = 2(K_{n=-4})^{1/2} P_{O2}^{-1/4} \quad (23)$$

Here, it was assumed that $[e']_{n=-4}$ is equal to $[e']_{Sto.}$ under P_{O_2}, and the following equations were obtained.

$$\Delta H_{n=-4} = \Delta H_V - \Delta H_F/2 \quad (24)$$

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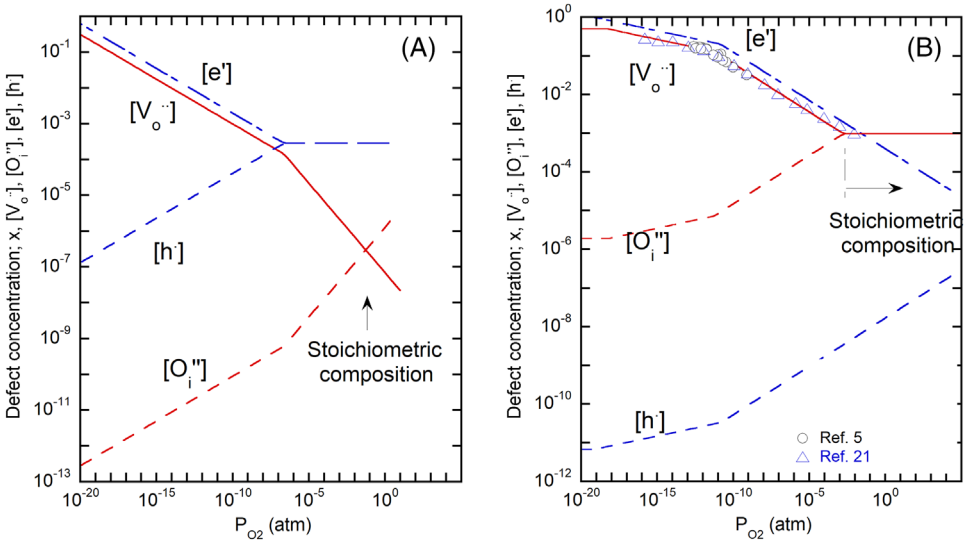

FIGURE 2 Brouwer diagram for (A) PuO_2 and (B) CeO_2 at 1473 K

TABLE 2 ΔH_m and ΔS_m of equilibrium constants for defect reactions

	ΔH_m (kJ/mol)	ΔS_m (J/mol•K)
K_i	212.0	-70.0
K_F	413.0	165.0
K_V	546.5	183.1
K_O	290.5	-158.1
$K_{n=-4}$	340.0	89.09
$K_{n=-12}$	345.0	79.78

TABLE 3 Defect formation energies (kJ/mol) in CeO_2 and PuO_2

	CeO_2	PuO_2[11]
Electron-hole pair	212.0	325.0
Frenkel defect	413.0	441.0
Oxygen vacancy	206.5	282.5
Interstitial oxygen	206.5	159.3
Electron	170.0	162.5
Hole	42.0	162.5

$$\Delta S_{n=-4} = \Delta S_V - \Delta S_F/2 - 2 \cdot R \cdot ln(2) \tag{25}$$

DFT calculation showed that the Frenkel defect formation energy and the bandgap should be 413 and 212 kJ/mol, and these data were input to ΔH_F and ΔH_i, respectively. ΔS_F and ΔS_i were determined to construct a Brouwer diagram in a temperature region of 1073–1773 K and all parameters were obtained as shown in Table 2. Figure 2 shows the Brouwer diagram at 1473 K, which describes the defect equilibria in CeO_2. The diagram represented the defect equilibria under P_{O_2}, and showed that extrinsic defects are dominant in the near-stoichiometric region and change to intrinsic defects in a P_{O_2} region of less than about 10^{-2} atm. It is known that dominant impurities such as lanthanide elements behave as acceptor dopants. Impurities having a valency of +3 form oxygen vacancy and reduce the upper limited Oxygen to Metal (O/M) ratio in CeO_2 from 2.00 depending on the content of impurities.

In Figure 2, the Brouwer diagram of PuO_2 is also shown for comparison. The figure shows that the P_{O_2} at exact stoichiometric composition is 4×10^{-2} atm and greater than 1.7×10^{-3}atm in PuO_2 and CeO_2, respectively, which are shown with arrows. Two types of Brouwer diagrams are proposed in which the Frenkel defect is dominant and the intrinsic defect is dominant at stoichiometric composition. The difference between the diagrams of CeO_2 and PuO_2 is the defect stability near stoichiometry composition, which corresponds to the two types of Brouwer diagram. In the case of CeO_2, $[Vo^{\cdot\cdot}]$ and $[Oi'']$ dominate $[e']$ and $[h^{\cdot}]$. On the other hand, $[e']$ and $[h^{\cdot}]$ are important in PuO_2. Formation energies of defects in both materials are compared in Table 3. The Frenkel defect formation energy of CeO_2 was evaluated as 413 kJ/mol. Beschnitt et al. reported that the energy was 399.0 kJ/mol (4.14 eV),[24] which is almost the same as the current data. It is also in good agreement with PuO_2 data. The electron-hole pair formation energy of CeO_2 was about 100 kJ/mol lower than that of PuO_2. To better understand the difference between the two Brower diagrams, it is necessary to investigate the behavior of lattice defects such as the Frenkel and electronic defects at high temperatures.

DFT calculation obtained that band gap energy was 212 kJ/mol (2.2 eV). The calculation result was about 100 kJ/mol lower as compared with other studies which

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were reported to be 289–338 kJ/mol (3–3.5 eV).[1,25,26] In this paper, the bandgap energy was assumed to be the formation energy of the electron-hole pair. If 289–338 kJ/mol was used as the bandgap energy, formation energy of $h^{\cdot}$ and $[h^{\cdot}]$ were affected, the formation energy of $h^{\cdot}$ became 119–168 kJ-mol and $[h^{\cdot}]$ was several orders of magnitude smaller. The evaluation result of hole formation affects properties in hyper-stoichiometric composition, but CeO_2 does not exist in the hyper-stoichiometric composition range. So, the effect of bandgap energy is negligibly small in the thermophysical property evaluation described below.

3.3 | Oxygen diffusion coefficient

Defect concentration can be calculated as a function of temperature and P_{O_2} using the Brouwer diagram constructed. P_{O_2} dependence on the defect concentration of $[Vo^{\cdot\cdot}]$ and $[Oi'']$ correlate strongly with the diffusion coefficients. In this section, the relationship between diffusion coefficients and defect concentrations of $[Vo^{\cdot\cdot}]$ and $[Oi'']$ is described.

The oxygen self-diffusion coefficient has been investigated as a function of temperature and P_{O_2}. The self-diffusion coefficient D^* was evaluated by the following equation.

$$D^* = D^O \times \exp\left(-\frac{E_{ion}}{RT}\right) \tag{26}$$

Here, E_{ion} is the activation energy, and D^O is a constant. The self-diffusion coefficient D^* and the chemical diffusion coefficient $\tilde{D}$ are written by

$$D^* = D^O_{Vo}\,[V_O{}^{\cdot\cdot}] \times \exp\left(-\frac{\Delta_m H_{VO}}{RT}\right) + 2 \times D^O_{Oi}\left[O_i{}''\right] \times \exp\left(-\frac{\Delta_m H_{Oi}}{RT}\right) \tag{27}$$

$$\tilde{D} = -\frac{2-x}{x} D^* \left(\frac{\partial logP_{O2}}{\partial logx}\right) \tag{28}$$

The obtained $\Delta_m H_{VO}$ and $\Delta_m H_{Oi}$ were 53.7 and 77 kJ/mol, respectively, by DFT calculation, as shown in Table 1. The values $[V_O{}^{\cdot\cdot}]$, $[O_i{}'']$,and $(\frac{\partial logP_{O2}}{\partial logx})$ can be evaluated by equations derived by constructing a Brouwer diagram. In a near-stoichiometric region, $[V_O{}^{\cdot\cdot}]$ and $[O_i{}'']$ are obtained by Equation (18). It is considered that oxygen vacancy diffusion is dominant in self-diffusion. The activation energy E can be described by,

$$E_{ion} = \Delta H_F/2 + \Delta_m H_{VO} \tag{29}$$

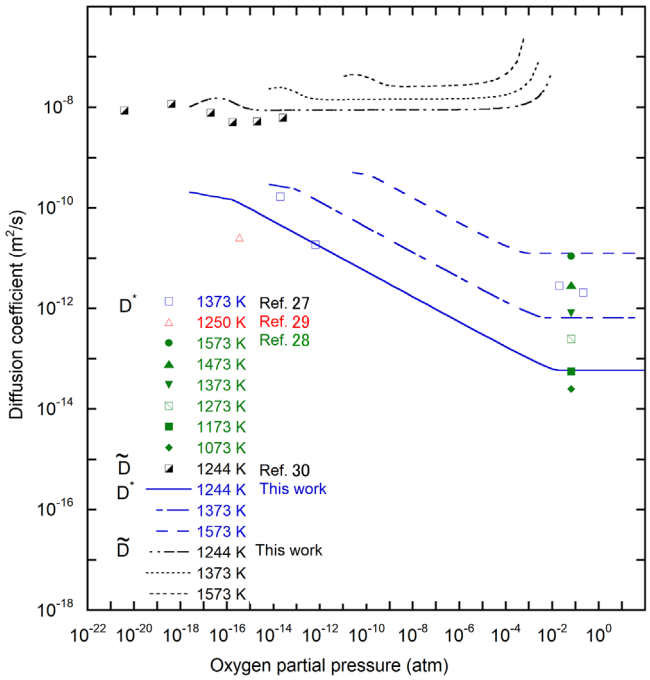

FIGURE 3 Oxygen partial pressure dependence of oxygen chemical and self-diffusion coefficients

The obtained ΔH_F and $\Delta_m H_{VO}$ were 413 and 53.7 kJ/mol, respectively, by DFT calculation. E was evaluated as 260.2 kJ/mol. Floyd[27] measured D^* in CeO_{2-x}, and E_{ion} for $CeO_{2.00}$ was determined to be 73–104 kJ/mol at temperatures of 1123–1423 K. Kamiya et al.[28] also measured D^* at $P_{O_2} = 0.07$ atm and reported that E_{ion} was 226 kJ/mol. Gotte et al.[29] evaluated D^* by molecular dynamic simulation as E_{ion} = 56 kJ/mol. The value evaluated by Kamiya et al.[28] was in good agreement with the current value, which was obtained by DFT simulation. Measurement of $\tilde{D}$ was carried out by Millot and Mierry[30] at 1244 K. The experimental data were plotted in Figure 3.

The P_{O_2} dependence of D^* and $\tilde{D}$ was calculated at 1244, 1373, and 1573 K by Equations (27) and (28) as shown in Figure 3. The calculation agreed well with the experimental data, when $D^O_{Vo} = 2 \times 10^{-7}$. D^* increases with decreasing P_{O_2}, which corresponds to the decrease in the O/M ratio.

3.4 | Electrical conduction

The electrical conductivity of oxides can be described by electronic and ionic conduction contributions, which are evaluated by defect concentrations. The electrical conductivity mechanism depends on P_{O_2} and was evaluated using $[e']$, $[h^{\cdot}]$, $[Vo^{\cdot\cdot}]$, and $[Oi'']$, which were calculated by the Brouwer diagram in this section.

The mechanism of electrical conduction was investigated.[6–9] It was reported that ion conduction

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and small polaron hopping conduction were dominant in a near-stoichiometric and a reduced region, respectively. Therefore, it was expected that electrical conductivity σ of CeO_{2-x} would be given by the sum,

$$\sigma_{total} = \sigma_{ion} + \sigma_{el} \tag{30}$$

Ionic and electronic-conductivities σ_{ion} and σ_{el} were written by

$$\sigma_{ion} = C_{ion/T} \times [Vo^{\cdot\cdot}] \times \exp\left(-\frac{\Delta_m H_V}{RT}\right) \tag{31}$$

and

$$\sigma_{el} = \sigma_n + \sigma_p = e\,[\mathrm{e}]\,\mu_n + e\,[\mathrm{h}]\,\mu_p = \frac{C_{\mu_e}}{T^{\frac{3}{2}}}[\mathrm{e}]\exp\left(\frac{E_{\mu_e}}{RT}\right) + \frac{C_{\mu_h}}{T^{\frac{3}{2}}}[\mathrm{h}]\exp\left(\frac{E_{\mu_h}}{RT}\right) \tag{32}$$

where C_{ion}, C_{μ_e},and C_{μ_h} are constants, and μ_e and μ_h are electron and hole mobilities, respectively. $[Vo^{\cdot\cdot}]$, [e], and [h] can be calculated as functions of temperature and P_{O_2} using Equations (18)–(23).

$$E_{ion} = \Delta H_F/2 + \Delta_m H_V \tag{33}$$

$$E_{n=-4} = \Delta H_{n=-4} + E_{\mu_e} \tag{34}$$

E_{ion} was obtained as 260.2 kJ/mol in self-diffusion evaluation using Equation (26). Chiang et al.[31] reported that E_{ion} was 236 kJ/mol (2.4 eV) in electrical conduction measurement. These values were in good agreement. Naik and Tien[7] and Blumenthal et al.[32] reported the P_{O_2} dependence of electrical conductivity of CeO_{2-x}. The data are plotted in Figure 4.[7] The total electrical conductivity was evaluated by Equations (31) and (32) assuming that $\mu_n = \mu_p$. Unknown parameters were determined by fitting the experimental data. Calculation results were shown in Figure 4. Analysis showed that E_{μ_e} = 80 kJ/mol, $C_{\mu_e} = 3 \times 10^6$, and $C_{\mu_e} = 1 \times 10^9$. It has been reported that E_{μ_e} is 20–50 kJ/mol for small polaron hopping conduction [6]. The value evaluated in this work was slightly higher compared with other data. Calculation results σ_{total}, σ_{ion}, and σ_{el} are shown in Figure 4, which agree with the experimental data well. In a reducing region of less than 10^{-1} atm, it was shown that σ_{total} was proportional to ${P_{O2}}^{-1/4}$. This relationship was consistent with the oxygen potential measurement data which were used to construct the Brouwer diagram in this work. Figure 4 shows the conduction mechanism in this region is observed as a mixed conduction region of σ_{ion} and σ_{el} in P_{O2} from 10^{-5} to 10^2 atm.

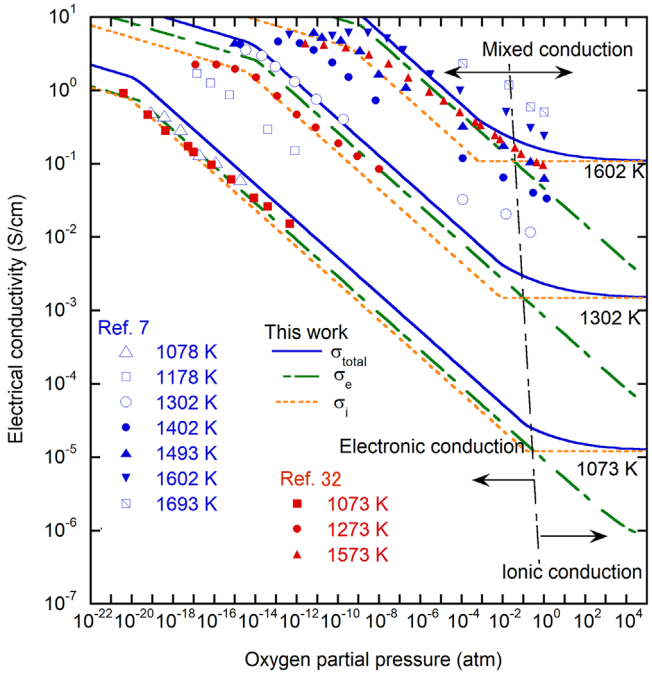

FIGURE 4 Oxygen partial pressure dependence of electrical conductivity

3.5 | Heat capacity

In this work, the mechanical and thermal properties of CeO_{2-x} were evaluated based on DFT simulation and oxygen potential experimental data. The heat capacity was represented using the evaluated data. The heat capacity C_p of oxides was evaluated using Equation (35)

$$C_p = C_l + C_d + C_e + C_f \tag{35}$$

Where C_l, C_d, C_e, and C_f are heat capacity terms related to lattice, dilatational, electronic defect formation and Frenkel defect formation, respectively. In the evaluation of C_p of PuO_2, the Schottky-type heat capacity was assumed. But, the Schottky-type heat capacity was not assumed in CeO_2 because of no *f* electrons.

C_l is calculated using the Debye model:

$$C_l = 9nR\left(\frac{T}{T_D}\right)^3 \int_0^{T_D/T} \frac{x^4 e^x}{(e^x - 1)^2} dx \tag{36}$$

The Debye temperature T_D can be written as:

$$T_D = \left(\frac{h}{k_B}\right)\left(\frac{9nR}{4 \cdot \pi \cdot a^3}\right)^{\frac{1}{3}}\left(\frac{1}{{V_l}^3} + \frac{2}{{V_t}^3}\right)^{-\frac{1}{3}} \tag{37}$$

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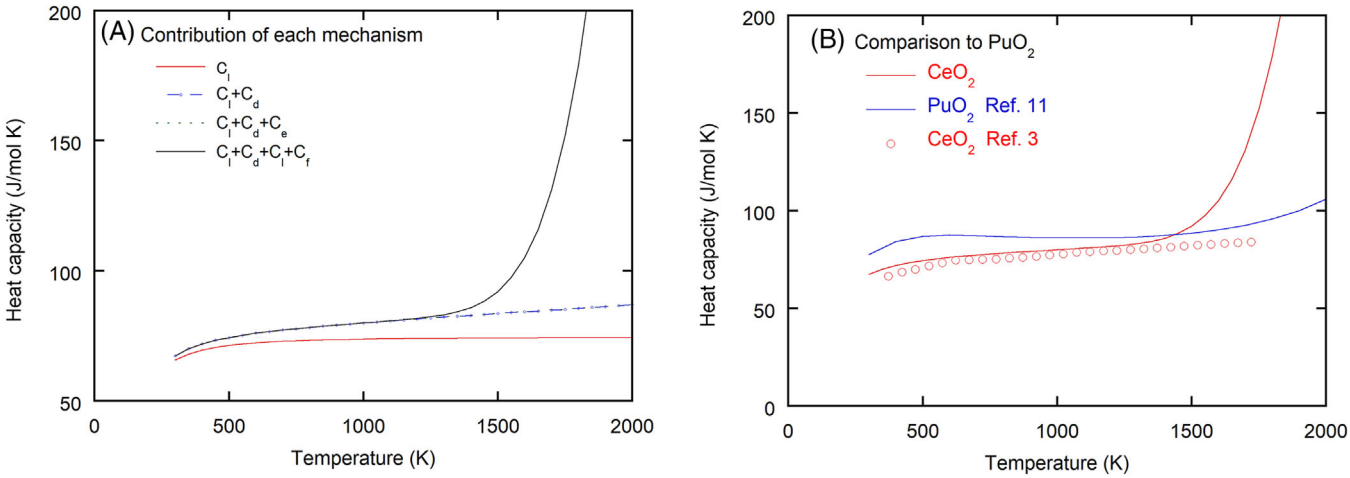

FIGURE 5 Temperature dependence of heat capacity

The value for C_d can be obtained as:

$$C_d = C_v \gamma \alpha T \tag{38}$$

The γ is the Grúneisen constant which is calculated by Equation (39):

$$\gamma = \alpha K V_m / C_v \tag{39}$$

Values of T_D, V_l, V_t, γ and α are listed in Table 1. T_D and γ of CeO_2 were obtained to be 489 K and 2.96, respectively. C_e and C_f can be described by Equations (40) and (41).

$$C_e = \frac{\Delta H_i^2}{2RT^2} \exp\left(\frac{-\Delta H_i}{2RT}\right) \exp\left(\frac{\Delta S_i}{2R}\right) \tag{40}$$

$$C_f = \frac{\Delta H_f^2}{\sqrt{2}RT^2} \exp\left(\frac{-\Delta H_f}{2RT}\right) \exp\left(\frac{\Delta S_f}{2R}\right) \tag{41}$$

ΔH and ΔS were already evaluated from the Brouwer diagram as shown in Table 2. Therefore, we can calculate the heat capacity of CeO_2, which is shown in Figure 5. The C_p of CeO_2 rapidly increased at temperatures of more than 1300 K. Figure 5 shows that the increase was caused by the effect of C_f and the effect of C_e was negligibly small. This suggests that the Frenkel defect concentration rapidly increases and disorder occurs at such high temperatures. It is expected that the Bredig transition occurs in the same way as UO_2 and CaF_2 at high temperatures.[33,34] In Figure 5B, C_p of CeO_2 is compared with other experimental data and that of PuO_2. The current analysis result was in good agreement with other experimental data of CeO_2, but reliability is unclear at high temperatures of more than 1500 K because of no experimental data at such high temperatures. The C_p of PuO_2 was higher than that of CeO_2 at temperatures lower than about 1400 K, which was caused by Schottky-type heat capacity. The C_p increase observed at higher temperatures in both oxides was caused by increased concentration of lattice defects which were electronic and Frenkel defects, respectively, in PuO_2 and CeO_2.

3.6 | Thermal conductivity

Thermal conductivity can be described by phonon and electrical conduction contributions, λ_p and λ_e, as follows.

$$\lambda = \lambda_p + \lambda_e = \frac{1}{BT} + (\Delta H_F/e)^2 \times \sigma/(4T) \tag{42}$$

Phonon conduction contribution λ_p was analyzed by Slack's equation[35] as follows:

$$\frac{1}{B} = 3.04 \times 10^7 \frac{\bar{M} {T_D}^3 \delta}{\gamma^2 n^{2/3}} \tag{43}$$

Values of T_D and γ are shown in Table 1. The average mass $\bar{M}$ is 172.1, the number of atoms per molecule$n$ is 3, and δ is $(V_m)^{1/3}$. V_m is average volume per atom, which was calculated by,

$$Vm = \frac{a^3}{4n} = 1.327 \times 10^{-29} m^3/atom \tag{44}$$

The elementary charge e is 96 400 C/mol. Electrical conductivity σ is obtained by Equation (31), and all parameters have been obtained to describe thermal conductivity. The calculation results are shown with the solid red line in Figure 6. The λ-value of CeO_2 is in good agreement with the previous data reported by Suzuki et al.,[5] which were corrected to full density and shown with the

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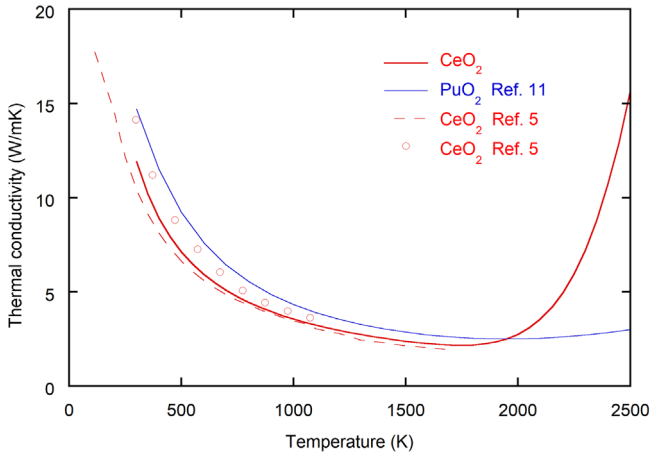

FIGURE 6 Temperature dependence of thermal conductivity

red broken line. Since the calculation results of Suzuki et al.[5] did not take the electrical conduction term into consideration, they did not increase at high temperatures. Although there are no experimental data on thermal conductivity and heat capacity at temperatures higher than 1600 K, the current evaluation results suggest that the thermal conductivity is increased by electrical conduction contribution.

4 | CONCLUSION

DFT simulation was employed to evaluate the properties of CeO_2 to determine bandgap, Frenkel defect formation energy, and defect migration energy. Defect equilibria were analyzed to construct a Brouwer diagram using the oxygen potential data, and the defect formation energies of electron-hole, Frenkel defect, oxygen vacancy, interstitial oxygen, electron, and hole were determined to be 212, 413, 206.5, 206.5, 170.0, and 42.0 kJ/mol, respectively. These evaluated data were applied to describe the oxygen diffusion coefficients, electrical conductivity, heat capacity, and thermal conductivity. Based on the defect equilibria, deviation x, oxygen diffusion coefficient, and electrical conductivity were evaluated as functions of P_{O_2} and temperature. In a near-stoichiometric region, the obtained activation energy of electrical conduction was 260.2 kJ/mol, which is in good agreement with the activation energy for the oxygen self-diffusion coefficient. This result supports the conclusion that stoichiometric CeO_2 is an ionic electrical conductor. In addition, electrical conduction in hypo-stoichiometric CeO_2 was evaluated as electronic conduction due to small polaron hopping conduction. Heat capacity and thermal conductivity of CeO_2 were also analyzed, and it was expected that both data would increase at high temperatures due to Frenkel defect concentration. The mechanisms of various properties were discussed, a better understanding was achieved based on defect chemistry, and the relationship among those properties was consistently and systematically evaluated.

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